



CIA2C14 Data Science Essentials

Data Preparation: Assess the Quality of the Data

What is Data Preparation?



Steps taken to shape
the data to the
desired quality

Can't run away from Data Wrangling



Most data is 'wild and messy'.
We need to **transform** them bring them under control.

DATA SET 1	DATA SET 2
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CUSTOMER	AMOUNT
Samsung Industries	200
Samsung	100
Pacific Industries	300
IBM	20
I.B.M.	40000
Samsung-Corning Solar Industries	130
Apple	200
Pacific Industries (Korea)	1200

10/22/2014
22/10/14
10/22/2014 1600
10/22/2014
41934



Common Problems

- I. Duplications
- II. Inconsistencies
- III. Missing values
- IV. Not in Tidy Format
- V. Not in correct data type

1001	Number
1002	Number
'1003'	Text
'1004'	Text

- Can one perform $1001 + 1002 + '1003' + '1004'$?

Spot errors with domain knowledge

- Use programming languages and tools to help us assess
- However 'error or not' sometimes require domain knowledge

☐ *160-year old person*

☐ *Negative Sales*

☐ *10,000 degree Celsius*

☐ *Local contact with 11 digits*

☐ *Correlated errors*

->X can be male

->X can be pregnant

->X cannot be male AND pregnant

Are these values data?

0

Zero values
are actual
data

<Blank>

Can be
whitespaces

NA

Not available
Is actual
data

NaN

Not A Number
**Missing value in
Python**

None

Is actual
data



Outcome is secondary, understanding why it happened is even more important

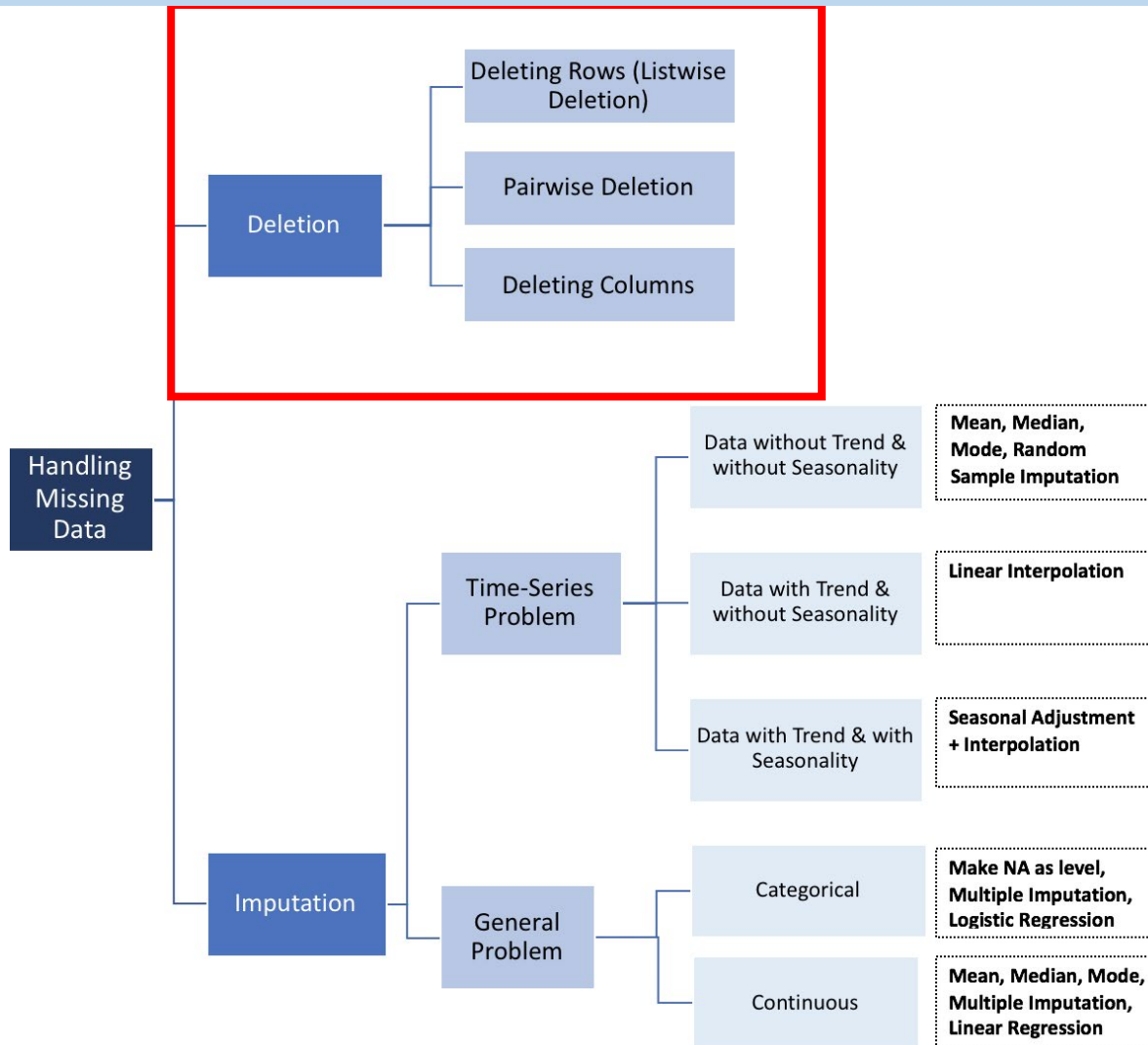
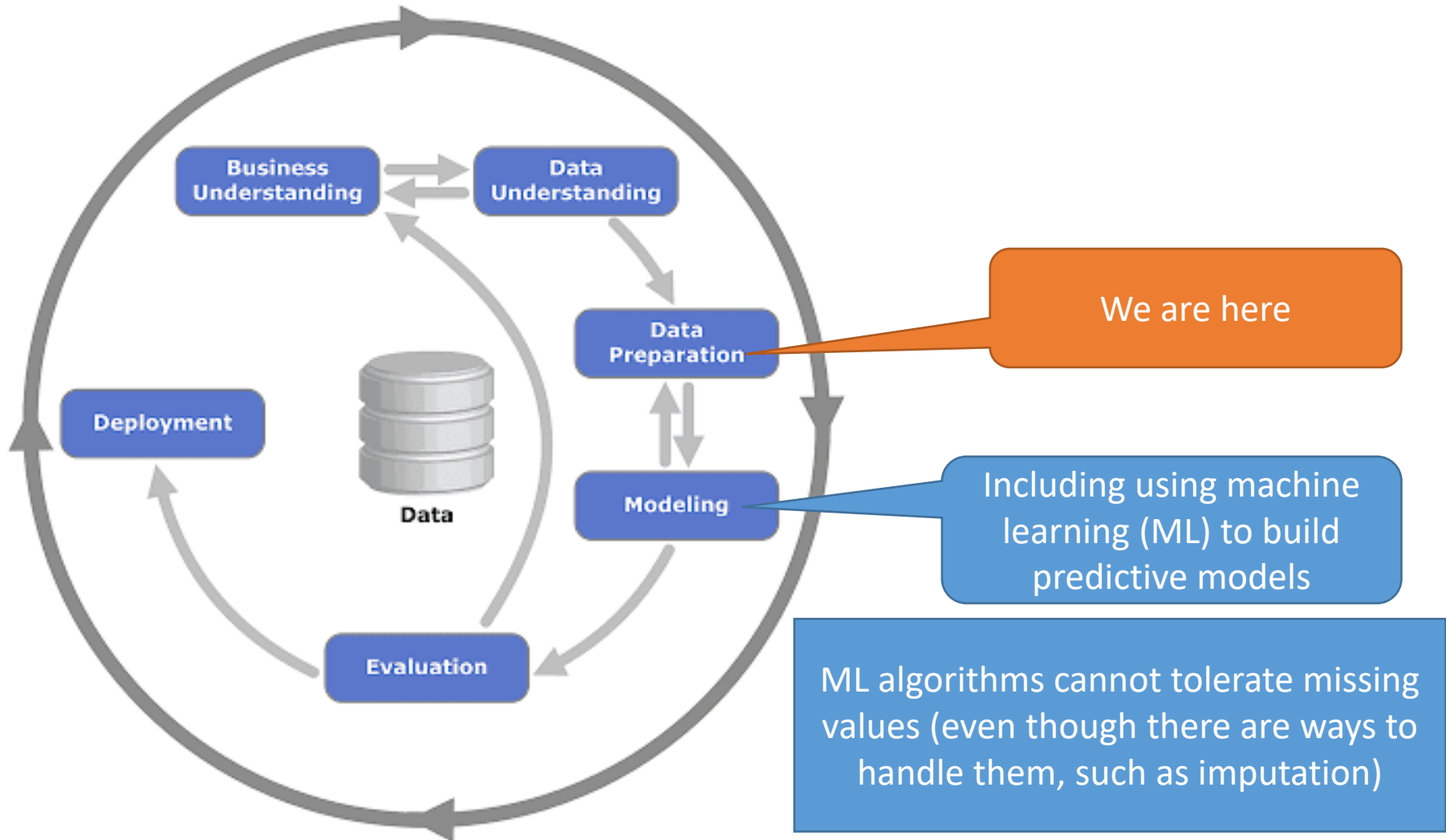


Image retrieved from:
<https://towardsdatascience.com/how-to-handle-missing-data-8646b18db0d4>

CRISP-DM



Commons steps involved

- i. **Missing values** detection (followed by imputation/deletion)
- ii. **Mathematical** transformations (e.g. take absolute values)
- iii. **Data-time** format (standard formatting is *YYYY-MM-DD*)
- iv. **Sort**
- v. Summarize (using statistical means) to **create new attributes** (columns)
- vi. **Merge/ Join** (from different data sources into one table)
- vii. Rename/ Encode (e.g. Male:0; Female: 1)
- viii. **Gather/ Spread** into Tidy Data format

The need for transformation



	A	B	C	D
1	Trainer	No. of Classes Conducted	Average Feedback	Session Dates
2	Trainer A	2 classes	3.5 / 5	3 Jan, 16 Jan
3	Trainer B	2 classes	3 / 5	4 Jan, 5 Jan
4	Trainer C	3 classes	4.7 / 5	7 Jan, 8 Jan, 9 Jan

Any quick insights derived from the new table?



	A	B	C
1	Trainer	Session	Feedback
2	A	3-Jan-17	5
3	A	3-Jan-17	5
4	A	3-Jan-17	5
5	A	16-Jan-17	2
6	A	16-Jan-17	2
7	A	16-Jan-17	2
8	B	4-Jan-17	3
9	B	4-Jan-17	0
10	B	4-Jan-17	4
11	B	4-Jan-17	4
12	B	5-Jan-17	3
13	B	5-Jan-17	3
14	B	5-Jan-17	3
15	B	5-Jan-17	4

...

**WE 
DATA
SCIENCE**